MILITARY SPECIFICATION

INDICATORS AND TRANSMITTERS, LIQUID,
TEMPERATURE AND PRESSURE

This amendment forms a part of Military Specification MIL-I-10986F, dated 11 September 1978, and is approved for use by the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

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*2.1, Specifications, Military, delete:

"MIL-E-13856 - Electrical Components for Automotive Vehicles; Waterproofness and Resistance to Temperature Cycling Tests, General Requirements for."

*2.1, Standards, Military, add.

"MIL-STD-1184 - Electrical Components for Automotive Vehicles; Waterproofness Tests."

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3.3.2.1.4, delete in its entirety and substitute:

"3.3.2.1.4 Cover window. The cover window shall be treated on both sides with an electrostatic solution, unless the window material has been tested for static charge generating characteristics and has been demonstrated to be nonsusceptible to a static charge that would deflect the pointer when the window is rubbed with a cloth containing not less than 95 percent fleece

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or pulled wool. The window shall be secured to the base so that moisture is excluded from the instrument case. The thickness of the window shall be 1/8 inch minimum. Window shall have an izod strength of not less than 0.3 foot pounds, and shall retain transparency with age. The window material shall conform to MIL-W-80 or, if molded, shall conform to type I, class 3 of L-P-380 or L-P-393."

3.3.2.2, line 2, delete "." and add ", or L-P-393 with 20 percent glass fiber added."

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*3.4.11.1, line 3, delete "waterproofness requirements" and substitute "failure criteria".

*3.4.11.1, line 4, delete "MIL-E-13856" and substitute "MIL-STD-1184."

*Add new paragraphs:

"3.4.11.1.1 Type III transmitters - waterproof test. When tested as specified in 4.5.13.1.1, type III transmitters shall show no evidence of leakage and shall subsequently meet the requirements of 3.4.10.2 for all calibration points on the pertaining MS drawing."

"3.4.14 Transient voltages. When tested as specified in 4.5.16 and 4.5.17, indicators shall show no evidence of damage or loss of calibration."

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*TABLE I, Unit 1 - Indicators, Specimen B, after "4.5.13.1 - Waterproofness", add: "4.5.16 - Positive transient voltage", and "4.5.17 - Negative transient voltage."

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*TABLE I, Unit 2 - Transmitters, Specimen C, between "4.5.13.1 (Types II and III) - Waterproofness" and "4.5.13.3 (Type I) - Waterproofness", add: "4.5.13.1.1 (Type III) - Waterproofness."

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*4.4.5.1, Item 2, after "Waterproofness 4.5.13.1 or 4.5.13.3" add: "Transient voltage 4.5.16 and 4.5.17".

*4.4.5.1 Item 3, after "Calibration 4.5.2 or 4.5.3" and prior to "Temperature influence 4.5.12" add: "Waterproofness 4.5.13.1.1".

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*4.5.13.1, line 3, after "III transmitters shall be subjected to the waterproofness test" add: ", Method 100,".

*4.5.13.1, line 4, delete "MIL-E-13856" and substitute "MIL-STD-1184."

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*Add new paragraph:

"4.5.13.1.1 Waterproofness, first article and control tests, type III transmitters. To determine conformance to 3.4.11.1.1, test type III transmitters in accordance with method 100, procedure 5 of MIL-STD-1184, except test pressure shall be 18 psi for 30 minutes. The threaded end shall be capped during the test."

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*4.5.13.3, line 3, delete "salt water solution of MIL-E-13856" and substitute "saline solution of MIL-STD-1184".

Add new paragraphs:

"4.5.16 Positive transient voltage. To determine conformance to 3.4.14, the indicator shall be connected in a test circuit as shown in figure 9. Subject the indicator to five pulses by cycling switch S1 from "B" to "A". Switch shall remain in each position for three minutes. The indicator shall then be tested in accordance with 4.5.2.

4.5.17 Negative transient voltage. To determine conformance to 3.4.14, the indicator shall be connected in a test circuit as shown in figure 10. Subject the indicator to five pulses by cycling switch S2 from "D" to "C". Switch shall remain in each position for one minute. The indicator shall then be tested in accordance with 4.5.2."

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Add Figures 9 and 10.

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Preparing activity:
Army - AT

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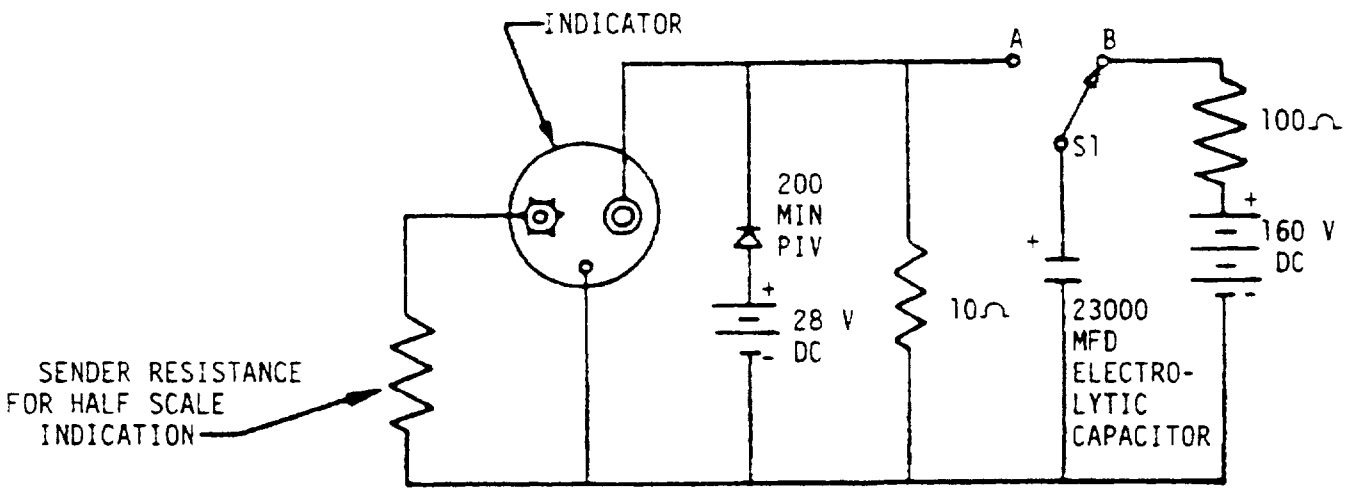


FIGURE 9. Positive transient test circuit.

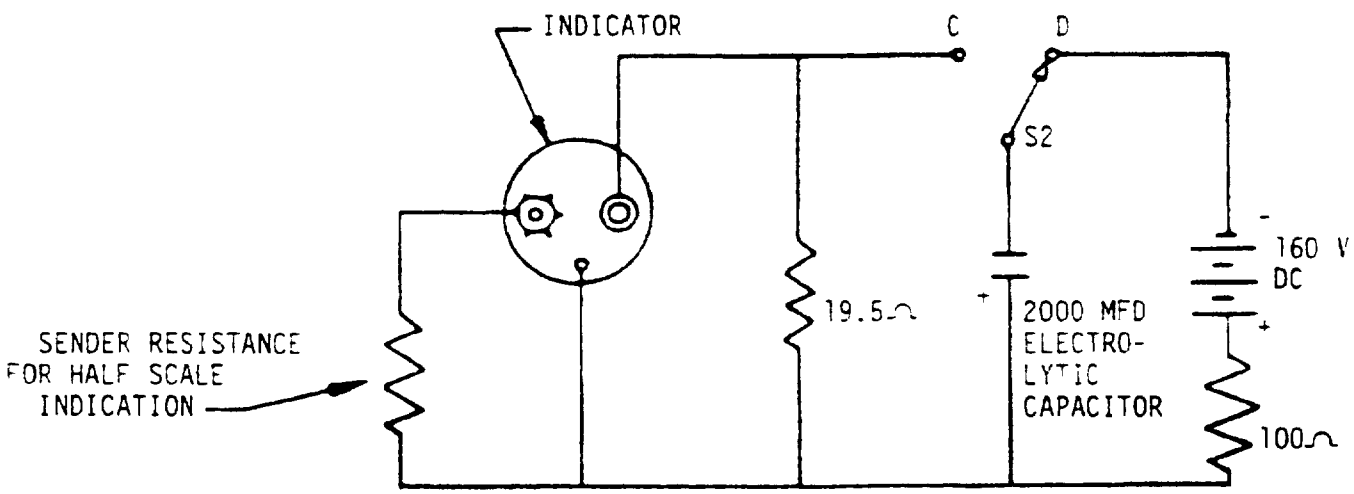


FIGURE 10. Negative transient test circuit.

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